Differential Geometry Excercise 4

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Question 1

Let $M \subset \mathbb{R}^3$ be a two-dimensional, embedded submanifold with the induced Riemannian metric. Let $U \subset \mathbb{R}^3$ be a neighborhood of 0 and let $f : \mathbb{R}^2 \to \mathbb{R}$ be a smooth function with $f(0) = 0$ and $\nabla f(0) = 0$. Let us assume that

$$M \cap U = \{(x, y, z) \in U : z = f(x, y)\} \tag{1}$$

Local orthonormal coframe

The induced metric comes from $\mathbb{R}^3$ Eucledian metric:

$$g_{\mathbb{R}^3} = dx^2 + dy^2 + dz^2 \tag{2}$$

Inside $M \cap U$, the induced metric takes the form:

$$g_{M\cap U} = dx^2 + dy^2 + (f_x dx + f_y dy)^2 = \left(1 + f_x^2\right) dx^2 + 2 f_x f_y dx dy + \left(1 + f_y^2\right) dy^2 \tag{3}$$

Where

$$f_x = \frac{\partial f}{\partial x} \qquad f_y = \frac{\partial f}{\partial y} \tag{4}$$

To find a coframe, let us use Cholesky decomposition. Define the two one-forms:

$$\omega_1 = a_x dx \qquad \omega_2 = b_x dx + b_y dy \tag{5}$$

Such that:

$$g = \omega_1^2 + \omega_2^2 \tag{6}$$

Openning this up gives:

$$g = (a_x^2 + b_x^2) dx^2 + 2 b_x b_y dx dy + b_y^2 dy^2 \tag{7}$$

comparing this to the original metric gives:

$$a_x = \sqrt{1 + \frac{f_x^2}{1 + f_y^2}} \qquad b_x = \frac{f_x f_y}{\sqrt{1 + f_y^2}} \qquad b_y = \sqrt{1 + f_y^2} \tag{8}$$

And so, and orthonotmal frame is:

$$\omega_1 = \sqrt{1 + \frac{f_x^2}{1 + f_y^2}} dx \qquad \omega_2 = \frac{f_x f_y}{\sqrt{1 + f_y^2}} dx + \sqrt{1 + f_y^2} dy \tag{9}$$

One form derivation

Let

$$\omega = \omega_1 + i\omega_2 \tag{10}$$

And let us look for the 1-form φ such that $d\omega = i\varphi \wedge \omega$. Plugging the defintion of ω into it gives:

$$d\omega_1 + i d\omega_2 = i\varphi \wedge (\omega_1 + i\omega_2) = -\varphi \wedge \omega_2 + i\varphi \wedge \omega_1 \tag{11}$$

Which requires:

$$\begin{cases} d\omega_1 = -\varphi \wedge \omega_2 \\ d\omega_2 = \varphi \wedge \omega_1 \end{cases} \tag{12}$$

Since ω_1, ω_2 is a coframe, φ can be written as its linear combination $\varphi = \varphi_1\omega_1 + \varphi_2\omega_2$:

$$\begin{cases} d\omega_1 = -\varphi_1 \omega_1 \wedge \omega_2 \\ d\omega_2 = -\varphi_2 \omega_1 \wedge \omega_2 \end{cases} \tag{13}$$

Let us now derive the values of these expressions in terms of dx and dy:

$$\omega_1 \wedge \omega_2 = \sqrt{\left(1 + \frac{f_x^2}{1+f_y^2}\right)\left(1+f_y^2\right)} dx \wedge dy = \sqrt{1+f_x^2+f_y^2} dx \wedge dy \tag{14}$$

$$\begin{aligned} d\omega_1 = -\frac{\partial a_x}{\partial y} dx \wedge dy = -a_x^{-1}\left(\frac{(1+f_y^2)f_x f_{xy} - f_x^2 f_y f_{yy}}{\left(1+f_y^2\right)^2}\right) dx \wedge dy = \\ = -\frac{f_x f_{xy} + f_y^2 f_x f_{xy} - f_x^2 f_y f_{yy}}{\left(1+f_y^2\right)^{3/2}\left(1+f_x^2+f_y^2\right)^{1/2}} dx \wedge dy = \\ = -\frac{f_x(f_{xy} + f_y^2 f_{xy} - f_x f_y f_{yy})}{\left(1+f_y^2\right)^{3/2}\left(1+f_x^2+f_y^2\right)^{1/2}} dx \wedge dy \end{aligned} \tag{15}$$

$$\begin{aligned} d\omega_2 = \left(\frac{\partial b_y}{\partial x} - \frac{\partial b_x}{\partial y}\right) dx \wedge dy = \\ = \left(\frac{f_y f_{xy}}{\sqrt{1+f_y^2}} - \frac{(f_{xy}f_y + f_x f_{yy})(1+f_y^2) - f_x f_y^2 f_{xy}}{(1+f_y^2)^{3/2}}\right) dx \wedge dy = \\ = \left(\frac{(f_{xy}f_y - f_x f_{yy} - f_y f_{xy})(1+f_y^2) + f_x f_y^2 f_{xy}}{(1+f_y^2)^{3/2}}\right) dx \wedge dy = \\ = -\frac{f_x(f_{yy} + f_y^2 f_{yy} + f_y^2 f_{xy})}{(1+f_y^2)^{3/2}} dx \wedge dy \end{aligned} \tag{16}$$

This gives:

$$\varphi_1 = \frac{f_x(f_{xy} + f_y^2 f_{xy} - f_x f_y f_{yy})}{\left(1+f_y^2\right)^{3/2}\left(1+f_x^2+f_y^2\right)} \qquad \varphi_2 = \frac{f_x(f_{yy} + f_y^2 f_{yy} + f_y^2 f_{xy})}{(1+f_y^2)^{3/2}\left(1+f_x^2+f_y^2\right)^{1/2}} \tag{17}$$

Theorema Egregium

Let us now proove Theorema Egregium at the origin, i.e. show that:

$$d\varphi = K\omega_1 \wedge \omega_2 \qquad s.t. \qquad K = \det \nabla^2 f(0) \tag{18}$$

To show that, let us expand f_x and f_y around the origin (all second derivatives from this point on are calculated at the origin, i.e. $f_{xx} = f_{xx}(0)$ and so forth):

$$f_x \approx f_{xx}x + f_{xy}y \qquad f_y \approx f_{xy}x + f_{yy}y \tag{19}$$

Pluggin that into the formulas for φ_1 and φ_2 and keeping only linear terms:

$$\varphi_1 = \frac{f_x(f_{xy} + f_y^2 f_{xy} - f_x f_y f_{yy})}{\left(1+f_y^2\right)^{3/2}\left(1+f_x^2+f_y^2\right)} \approx f_{xy}(f_{xx}x + f_{xy}y) \tag{20}$$

$$\varphi_2 = \frac{f_x(f_{yy} + f_y^2 f_{yy} + f_y^2 f_{xy})}{(1+f_y^2)^{3/2}\left(1+f_x^2+f_y^2\right)^{1/2}} \approx f_{yy}(f_{xx}x + f_{xy}y) \tag{21}$$

Which gives the form:

$$\begin{gathered}\varphi = f_{xy}(f_{xx}x + f_{xy}y)\omega_1 + f_{yy}(f_{xx}x + f_{xy}y)\omega_2 = \\ = (f_{xx}x + f_{xy}y)\left(f_{yy}\frac{f_x f_y}{\sqrt{1+f_y^2}} + f_{xy}\sqrt{1+\frac{f_x^2}{1+f_y^2}}\right)dx + f_{yy}(f_{xx}x + f_{xy}y)\sqrt{1+f_y^2}dy \approx \\ \approx f_{xy}(f_{xx}x + f_{xy}y)dx + f_{yy}(f_{xx}x + f_{xy}y)dy\end{gathered} \tag{22}$$

Deriving this, we get:

$$\begin{gathered}d\varphi = \left(\frac{\partial}{\partial x}\left(f_{yy}(f_{xx}x + f_{xy}y)\right) - \frac{\partial}{\partial y}\left(f_{xy}(f_{xx}x + f_{xy}y)\right)\right)dx \wedge dy = \\ = \left(f_{xx}f_{yy} - f_{xy}^2\right)dx \wedge dy = \det \nabla^2 f(0) dx \wedge dy\end{gathered} \tag{23}$$

But, in the origin:

$$\omega_1 \wedge \omega_2 = \sqrt{1+f_x^2+f_y^2}dx \wedge dy = dx \wedge dy \tag{24}$$

Thus:

$$d\varphi = \det \nabla^2 f(0)\omega_1 \wedge \omega_2 \tag{25}$$

Question 2

The gaussian curvature in $\mathbb{H}^2$

Let $\mathbb{H}^2$ be the upper half plane with the metric:

$$g = \frac{dx^2 + dy^2}{y^2} \tag{26}$$

Let us do Gram-Schmidt to find an orthonormal frame E_1, E_2. Starting from

$$E_1 = \partial_x \tag{27}$$

And then:

$$E_2 = \partial_y - \frac{y^{-2}\left(0\cdot 1 + 1 \cdot 0\right)}{y^{-2}}\partial_x = \partial_y \tag{28}$$

So ∂_x, ∂_y are already an orthogonal frame. Normalizing it gives:

$$E_1 = y\partial_x \qquad E_2 = y\partial_y \tag{29}$$

and so and orthonormal coframe is:

$$\omega_1 = y^{-1}dx \qquad \omega_2 = y^{-1}dy \tag{30}$$

And so

$$d\omega_1 = y^{-2}dx \wedge dy \qquad d\omega_2 = 0 \tag{31}$$

So we can concluate that:

$$\varphi = -\omega_1 \tag{32}$$

Since then:

$$d\omega_1 = -\varphi \wedge \omega_2 = \omega_1 \wedge \omega_2 = y^{-2}dx \wedge dy \qquad d\omega_2 = \varphi \wedge \omega_1 = -\omega_1 \wedge \omega_1 = 0 \tag{33}$$

Now, since

$$\omega_1 \wedge \omega_2 = y^{-2}dx \wedge dy \tag{34}$$

then:

$$d\varphi = -d\omega_1 = -y^{-2}dx \wedge dy = -\omega_1 \wedge \omega_2 \tag{35}$$

And so:

$$K = -1 \tag{36}$$

The Sphere S^2

For the sphere. Using the stereographic coordinate system, the metric is:

$$g = \frac{4}{(1+x^2+y^2)^2}\left(dx^2 + dy^2\right) \tag{37}$$

Similar to the half-plane, this gives the orthonormal frame:

$$E_1 = \frac{1+x^2+y^2}{2}\partial_x \qquad E_2 = \frac{1+x^2+y^2}{2}\partial_y \tag{38}$$

And the orthonormal coframe:

$$\omega_1 = \frac{2}{1+x^2+y^2}dx \qquad \omega_2 = \frac{2}{1+x^2+y^2}dy \tag{39}$$

the derivatives are then:

$$\begin{aligned} d\omega_1 &= -\frac{4ydy}{\left(1+x^2+y^2\right)^2} \wedge dx = \frac{4ydx}{\left(1+x^2+y^2\right)^2} \wedge dy = \frac{2y}{1+x^2+y^2}dx \wedge d\omega_2 \\ d\omega_2 &= -\frac{4xdx}{\left(1+x^2+y^2\right)^2} \wedge dy = \frac{4xdy}{\left(1+x^2+y^2\right)^2} \wedge dx = \frac{2x}{1+x^2+y^2}dy \wedge d\omega_1 \end{aligned} \tag{40}$$

And the cannoincal connection form is:

$$\varphi = \frac{2xdy - 2ydx}{1+x^2+y^2} \tag{41}$$

Deriving it:

$$d\varphi = \frac{4}{1+x^2+y^2}dx \wedge dy \tag{42}$$

On the other hand

$$\omega_1 \wedge \omega_2 = \frac{4}{1+x^2+y^2}dx \wedge dy \tag{43}$$

So

$$K = 1 \tag{44}$$

The Flat Torus

Let us look at the torus with the flat metric $T = \mathbb{R}^2/\mathbb{Z}^2$. This Torus has the same metric as $\mathbb{R}^2$, and is therefore flat. Let us show it. Starting from the metric:

$$g = dx^2 + dy^2 \tag{45}$$

This gives:

$$\omega_1 = dx \qquad \omega_2 = dy \tag{46}$$

Then:

$$d\omega_1 = d\omega_2 = 0 \tag{47}$$

And

$$\varphi = 0 \tag{48}$$

Therefore:

$$K = 0 \tag{49}$$

Question 3

Let M be a smooth manifold. Let $\{U_i\}$ be a cover of M and $\{\psi\}$ a partition of unity over M subordinate to this cover. Then, for every subset U_i define a smooth orthonormal coframe $\omega_1, \ldots, \omega_n$ on U_i. This gives a locally defined (0,2)-tensor field:

$$g_i = \sum_j^n \omega_j \otimes \omega_j \tag{50}$$

that is symmetric and positive. The global (0,2)-tensor field

$$g = \sum_i \psi_i g_i \tag{51}$$

is smooth and thus a metric.
Now, let $(p, X) \in TM$ be a member of the tangent bundle over M and define $E_1, \ldots, E_n$ as the orthonormal frame dual to ω_i:

$$E_i \omega_j = \delta_{ij} \tag{52}$$

Then, there exist $a_1, \ldots, a_n$ such that

$$X = \sum_i a_i E_i \tag{53}$$

It's easy to see that

$$g(X, E_i) = g(\sum a_i E_i, E_i) = \sum a_i g(E_i, E_i) = a_i \tag{54}$$

And since g and E_i are smooth, that means a_i are smooth as well. So, define the bijection $f : TM \to T^*M$ as the function the for each (p, X) gives

$$f(p, X) = \sum_i a_i \omega_i \tag{55}$$

Since a_i are smooth and ω_i are smooth, this is a smooth bijection with a smooth inverse (defined the same way), and therefore a diffeomorphism.

Question 4

Let $M \subset \mathbb{R}^n$ be a $n-1$-dimensional smooth manifold and let $\omega = dx_1 \wedge \cdots \wedge dx_n$ be the standard volume form on $\mathbb{R}^n$. Let N be the unit normal vector field. Then $\iota_N \omega$ is a $n-1$-form over M. Define an orthonormal frame $e_1, \ldots, e_{n-1}$ over M. Since $\langle N, X \rangle = 0$ for every $X \in TM$, the set $N, e_1, \ldots, e_{n-1}$ is orthonormal and thus a basis for $\mathbb{R}^n$. Then:

$$\iota_N \omega(e_1, \ldots, e_{n-1}) = \omega(N, e_1, \ldots, e_{n-1}) = \pm 1 \tag{56}$$

Which means $\iota_N \omega$ is a Riemannian volume form on M.

Question 5

Let $X, Y, X \in \Gamma(TM)$ be vector fields and let ∇ be a torsion free metric compatible connection. Then, because it is metric compatible:

$$\begin{aligned} Xg(Y,Z) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \\ Yg(Z,X) &= g(\nabla_Y Z, X) + g(Z, \nabla_Y X) \\ Zg(X,Y) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y) \end{aligned} \tag{57}$$

Adding the first two equations and substracting the last one:

$$\begin{gathered} Xg(Y,Z) + Yg(Z,X) + Zg(X,Y) = \\ = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) + g(\nabla_Y Z, X) + g(Z, \nabla_Y X) - g(\nabla_Z X, Y) - g(X, \nabla_Z Y) \end{gathered} \tag{58}$$

But since ∇ is torsion free:

$$\begin{aligned} g(Z, \nabla_Y X) &= g(Z, \nabla_X Y) - g(Z, [X,Y]) \\ g(Y, \nabla_X Z) - g(\nabla_Z X, Y) &= g(Y, [X,Z]) \\ g(\nabla_Y Z, X) - g(X, \nabla_Z Y) &= g(X, [Y,Z]) \end{aligned} \tag{59}$$

This gives:

$$Xg(Y,Z) + Yg(Z,X) - Zg(X,Y) = 2g(Z, \nabla_X Y) - g(Z, [X,Y]) + g(Y, [X,Z]) + g(X, [Y,Z]) \tag{60}$$

Reogranizing:

$$2g(Z, \nabla_X Y) = Xg(Y,Z) + Yg(Z,X) - Zg(X,Y) + g(Z, [X,Y]) - g(Y, [X,Z]) - g(X, [Y,Z]) \tag{61}$$

[1]

Question 6

Let V be a vector field along a curve $\gamma(t)$ and let $\tilde{V}_1$ and $\tilde{V}_2$ be two extensions of the vector field to a small neighborhood around the curve. Then, for every connection ∇, using the linearity of the connection:

$$\nabla_{\dot{\gamma}(t)} \left(\tilde{V}_1 - \tilde{V}_2 \right) = \nabla_{\dot{\gamma}(t)} \tilde{V}_1 - \nabla_{\dot{\gamma}(t)} \tilde{V}_2 \tag{62}$$

But since along the curve $\tilde{V}_1 = \tilde{V}_2 = V$, then $\tilde{V}_1 - \tilde{V}_2 = 0$ which gives

$$\nabla_{\dot{\gamma}(t)} \tilde{V}_1 - \nabla_{\dot{\gamma}(t)} \tilde{V}_2 = 0 \tag{63}$$

$$\nabla_{\dot{\gamma}(t)} \tilde{V}_1 = \nabla_{\dot{\gamma}(t)} \tilde{V}_2 \tag{64}$$

But since this is just the covariant derivative of V, it means the covariant derivation is independent of the extension.

[1] The worksheet has a wrong sign here, but I double checked and this is the formulation. I actually didn't notice it until I did Q11, where terms vanished with this formula wrong

Question 7

Let ∇ be the Levi Civita connection, i.e. the torsion free metric compatible connection. Then, define a local coordinate system $x_1, \ldots, x_n$ for every point $p \in M$ and a local frame $\partial_{x_1}, \ldots, \partial_{x_n}$. Then, the Christoffel symbols are

$$\nabla_{\partial_i} \partial_j = \sum_n \Gamma^n_{ij} \partial_n \tag{65}$$

Applying the metric with ∂_l:

$$g\left(\partial_l, \nabla_{\partial_i} \partial_j\right) = g\left(\partial_l, \sum_n \Gamma^n_{ij} \partial_n\right) \tag{66}$$

The RHS gives:

$$g\left(\partial_l, \sum_n \Gamma^n_{ij} \partial_n\right) = \sum_n \Gamma^n_{ij} g(\partial_l, \partial_n) = g_{ln} \Gamma^n_{ij} \tag{67}$$

Where the last step is written in Einstein summation convention. Multiplying by the inverse metric g^{lk} and summing over l gives:

$$g^{lk} g_{ln} \Gamma^n_{ij} = \delta_{kn} \Gamma^n_{ij} = \Gamma^k_{ij} \tag{68}$$

And so:

$$\Gamma^k_{ij} = g^{lk} g\left(\partial_l, \nabla_{\partial_i} \partial_j\right) \tag{69}$$

Using Koszul formula that we proved in Question 5:

$$\Gamma^k_{ij} = \frac{1}{2} g^{lk} \left[\partial_i g(\partial_j, \partial_l) + \partial_j g(\partial_l, \partial_i) - \partial_l g(\partial_i, \partial_j) + g(\partial_l, [\partial_i, \partial_j]) - g(\partial_j, [\partial_i, \partial_l]) - g(\partial_i, [\partial_j, \partial_l])\right] \tag{70}$$

But since $[\partial_i, \partial_j] = 0$ for every i, j:

$$\Gamma^k_{ij} = \frac{1}{2} g^{lk} \left[\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\right] \tag{71}$$

Question 8

Let $M = S^2$ be isometrically embedded in $\mathbb{R}^3$ and let g be the Eucledian metric. Now, let $\gamma(t)$ be the equator. In $\mathbb{R}^3$ Eucledian coordinates:

$$\gamma(t) = (\cos 2\pi t, \sin 2\pi t, 0) \tag{72}$$

Now, let $X = \partial_x$ be a parallel vector field in $\mathbb{R}^3$ and let Y be its orthogonal projection. Then on $\gamma(t)$ it can calculted by projecting on the tangent to the unit circle and then back to the Eucledian coordinates:

$$Y|_{\gamma(t)} = -\sin 2\pi t (\sin 2\pi t, \cos 2\pi t, 0) \tag{73}$$

So, for $t = 0$, $t = 0.125$ and $t = 0.25$ the projected vector field is:

$$Y|_{\gamma(0)} = (0, 0, 0) \qquad Y|_{\gamma(0.125)} = (0.5, 0.5, 0) \qquad Y|_{\gamma(0.25)} = (1, 0, 0) \tag{74}$$

So, this is an example of an orthogonal projection of a parall vector field that is not parallel in an isometrically embedded manifold in $\mathbb{R}^n$.

Question 11

Let (M, g) be an oriented Riemannian surface and (e_1, e_2) a local oriented orthonormal frame. Let $X \in TM$ be a vector field and let ∇ be the Levi Civita connection. writing X in terms of (e_1, e_2):

$$X = x_1 e_1 + x_2 e_2 \tag{75}$$

Gives:

$$\nabla_X e_1 = x_1 \nabla_{e_1} e_1 + x_2 \nabla_{e_2} e_1 \qquad \nabla_X e_2 = x_1 \nabla_{e_1} e_2 + x_2 \nabla_{e_2} e_2 \tag{76}$$

The object $\nabla_{e_i} e_j$ can be calcualted by applying the metric with e_k to it, and using Koszul formula:

$$g(\nabla_{e_i} e_j, e_k) = \frac{1}{2}\left[e_i g(e_j, e_k) + e_j g(e_k, e_i) - e_k g(e_i, e_j) + g(e_k, [e_i, e_j]) - g(e_j, [e_i, e_k]) - g(e_i, [e_j, e_k])\right] \tag{77}$$

And using $e_i g(e_j, e_k) = e_i g_{jk} = e_i \delta_{jk} = 0$:

$$g(\nabla_{e_i} e_j, e_k) = \frac{1}{2}\left[g(e_k, [e_i, e_j]) - g(e_j, [e_i, e_k]) - g(e_i, [e_j, e_k])\right] \tag{78}$$

Writing:

$$\nabla_{e_i} e_j = a_{ij} e_1 + b_{ij} e_2 \tag{79}$$

Gives:

$$a_{ij} = g(\nabla_{e_i} e_j, e_1) \qquad b_{ij} = g(\nabla_{e_i} e_j, e_2) \tag{80}$$

Specifically:

$$\begin{aligned}
a_{11} = g(\nabla_{e_1} e_1, e_1) &= \frac{1}{2}\left[g(e_1, [e_1, e_1]) - g(e_1, [e_1, e_1]) - g(e_1, [e_1, e_1])\right] = 0 \\
a_{12} = g(\nabla_{e_1} e_2, e_1) &= \frac{1}{2}\left[g(e_1, [e_1, e_2]) - g(e_2, [e_1, e_1]) - g(e_1, [e_2, e_1])\right] = g(e_1, [e_1, e_2]) \\
a_{21} = g(\nabla_{e_2} e_1, e_1) &= \frac{1}{2}\left[g(e_1, [e_2, e_1]) - g(e_1, [e_2, e_1]) - g(e_2, [e_1, e_1])\right] = 0 \\
a_{22} = g(\nabla_{e_2} e_2, e_1) &= \frac{1}{2}\left[g(e_1, [e_2, e_2]) - g(e_2, [e_2, e_1]) - g(e_2, [e_2, e_1])\right] = g(e_2, [e_1, e_2])
\end{aligned} \tag{81}$$

And similarly:

$$b_{12} = b_{22} = 0 \qquad b_{11} = g(e_1, [e_2, e_1]) \qquad b_{21} = g(e_2, [e_2, e_1]) \tag{82}$$

So, plugging those into the above formulation:

$$\nabla_{e_1} e_1 = g(e_1, [e_2, e_1]) e_2 \qquad \nabla_{e_1} e_2 = g(e_1, [e_1, e_2]) e_1 \qquad \nabla_{e_2} e_1 = g(e_2, [e_2, e_1]) e_2 \qquad \nabla_{e_2} e_2 = g(e_2, [e_1, e_2]) e_1 \tag{83}$$

And:

$$\begin{aligned}
\nabla_X e_1 &= x_1 g(e_1, [e_2, e_1]) e_2 + x_2 g(e_2, [e_2, e_1]) e_2 \\
\nabla_X e_2 &= x_1 g(e_1, [e_1, e_2]) e_1 + x_2 g(e_2, [e_1, e_2]) e_1
\end{aligned} \tag{84}$$

So, defining the form

$$\omega = g(e_1, [e_2, e_1]) \omega_1 + g(e_2, [e_2, e_1]) \omega_2 \tag{85}$$

where ω_1, ω_2 be the coframe dual to e_1, e_2 gives:

$$\nabla e_1 = \omega \otimes e_2 \qquad \nabla e_2 = -\omega \otimes e_1 \tag{86}$$

This is a very similar structure to the cannoical connection φ. Then there exist a 1-form φ such that:

$$d\omega_1 = \varphi \wedge \omega_2 \qquad d\omega_2 = -\varphi \wedge \omega_1 \tag{87}$$

And then

$$d\varphi = K\omega_1 \wedge \omega_2 \tag{88}$$

But we notice that:

$$\omega \wedge \omega_2 = g(e_1, [e_2, e_1])\omega_1 \wedge \omega_2 \tag{89}$$

But also, since $e_j(\omega_1 e_i) = e_j \delta_{i1} = 0$:

$$d\omega_1(e_i, e_j) = e_i\omega_1 e_j + e_j\omega_1 e_i - \omega_1([e_i, e_j]) = \omega_1([e_j, e_i]) \tag{90}$$

Which gives $d\omega_1(e_1, e_1) = d\omega_1(e_2, e_2) = 0$ and $d\omega_1(e_1, e_2) = -d\omega_1(e_2, e_1) = \omega_1([e_2, e_1]) = g(e_1, [e_2, e_1])$. This gives:

$$d\omega_1 = g(e_1, [e_2, e_1])\omega_1 \wedge \omega_2 = \omega \wedge \omega_2 \tag{91}$$

A similar calculation for $\omega \wedge \omega_1$ and $d\omega_2$ will give the opposite equation. So, ω follows the first structure equation, and thus $\omega = \varphi$ and

$$d\omega = K\omega_1 \wedge \omega_2 \tag{92}$$